

Assignment I – NS (CSC4065)

Deadline: 09/03/2026 11:59 PM

This exercise is based on a real-world protocol (with minor modifications/simplifications), it received a thorough security analysis, and had attacks that the analysts missed. Have fun finding and fixing them! The exercise is accompanied with an AnB file on GCR.

NOTE: PLEASE BE CONSCISE IN YOUR ANSWERS

1. Before analyzing the attack, try to understand what purpose this protocol tries to achieve. To that end, first look at the initial knowledge and the goals. Can you give a typical real-world scenario from the Internet where the participants have this initial knowledge and want to achieve this goal?
2. Describe the attack found by OFMC by running:

ofmc task.AnB

What does the attacker do and what goal is broken here?

3. How could this attack be prevented for two sessions¹ without changing the initial knowledge of the roles or the goals (or using the channel notation)? To answer that it can help to imagine what the protocol designers had in mind with the different messages of the protocol and where the actual attack deviates from their idea: possibly there is just a small modification to fix the problem.
4. If you add the following goal:

sk(A,s) guessable secret between A,s

then OFMC finds another attack against the original protocol, and probably also against your fix (depending on how your fix works). Explain why a guessable $sk(A,s)$ can be a problem and try to find a fix for that as well.

¹ To run OFMC for two sessions on the protocol, simply add the following:
ofmc task.AnB -numSess 2